

SATHWIK H NAIK

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PROFESSIONAL SUMMARY

Recent M.S. Data Science graduate (University of Maryland, College Park, 2026, GPA 3.81/4.0) with ~1 year of corporate analytics experience at a Fortune 50 healthcare organization and hands-on depth across data engineering, data analytics, AI/RAG engineering, and data science. Built Power BI dashboards, ETL pipelines, and workflow automation at Elevance Health, and completed eight self-directed capstones spanning Kafka/Spark/dbt/Snowflake lakehouse pipelines, AWS-style streaming data reliability engineering, SQL/BI analytics, RAG system evaluation, banking forecasting and causal inference, and generative and small-language-model research.

EDUCATION

University of Maryland, College Park <i>M.S. in Data Science</i>	2024 – 2026 GPA 3.81/4.0
Coursework Statistical Inference, Deep Learning, Data Modeling, Data Visualization, Experimental Design, Advanced Time Series	
Reva University, India <i>B.Tech, Computer Science and Engineering</i>	2019 – 2023 GPA 3.55/4.0

WORK EXPERIENCE

Elevance Health <i>Software Engineer Fortune 50 healthcare organization</i>	Aug 2023 – Aug 2024
Tech Stack: Salesforce CRM, Informatica, Bitbucket, Power BI, DAX, AutoRABIT, Apex	
- Designed reusable KPI structures, translating operational requirements into consistent, self-serve cross-team metrics - Built and maintained 10+ Power BI dashboards with DAX measures for multiple internal teams - Developed ETL workflows to source, clean, and validate healthcare data, improving quality for stakeholder reports - Automated recurring data-preparation and reporting workflows, eliminating 20+ hours of manual weekly effort - Documented report logic, metric definitions, and validation rules enabling governed self-service reporting - Supported CI/CD pipelines for Salesforce releases using automated logging and standardized deployment workflows	

PROJECTS

Healthcare Insurance Claims Analytics & Fraud Signal Platform ([GitHub](#))

Tech Stack: Kafka (Redpanda), Spark (Structured Streaming), Delta Lake/Databricks, dbt, Airflow, Snowflake, Terraform

- Built a hybrid lakehouse-warehouse pipeline using **Kafka**, **Spark**, **Delta Lake**, **dbt**, **Airflow**, and **Snowflake**
- Implemented **Bronze/Silver/Gold medallion architecture** on Delta Lake processing 50,000 synthetic insurance claims
- Built two **dbt** projects with tests, transforming raw claims into governed, BI-ready **Snowflake** marts
- Provisioned **Databricks** and **Snowflake** infrastructure declaratively using **Terraform** with key-pair JWT authentication
- Built a rolling-baseline **fraud detection** rule in dbt achieving **100% recall and 67% precision**
- Built **Spark Structured Streaming** ingestion using explicit schemas, checkpointing, and micro-batch triggers

AI Support Telemetry Reliability Platform ([GitHub](#))

Tech Stack: TypeScript, Node.js, Amazon Kinesis (LocalStack), Amazon S3, AWS CDK, Ajv/JSON Schema, DuckDB

- Built a **TypeScript** streaming telemetry platform using **Kinesis-compatible APIs**, **LocalStack**, **S3**, and **DuckDB**
- Reconciled **32,154** synthetic transport records to **zero unaccounted loss** across accepted and quarantine zones
- Designed at-least-once canonicalization removing **600** injected duplicates with zero duplicate canonical keys
- Enforced versioned **JSON Schema** contracts via **Ajv**, isolating 50/50 invalid records at **100% precision and recall**
- Synthesized a least-privilege **AWS CDK** stack with **Lambda** partial-batch handling and **CloudWatch** alarms
- Drained a backlog from a 60-second simulated consumer outage in **1.748 seconds** after resuming

SaaS Subscription Churn & Revenue Analytics Platform ([GitHub](#))

Tech Stack: Python (pandas, NumPy), DuckDB, PostgreSQL, Grafana (config-as-code), Plotly, scikit-learn, SHAP

- Built a SaaS churn analytics platform across **DuckDB**, **pandas**, **PostgreSQL**, **Grafana**, and **scikit-learn**
- Modeled the **MRR waterfall** and running balance in SQL using CTEs and window functions
- Reconstructed point-in-time customer state using pandas **merge_asof** across 8,000 synthetic customer accounts
- Derived 15+ metrics tables including **NR**, **GRR**, and **ARPA** from raw subscription events
- Built a **21-panel Grafana** dashboard provisioned entirely as version-controlled config, not UI clicks
- Trained a **Gradient Boosting** churn model and explained predictions per-customer using **SHAP**

RAG Evaluation Lab ("Retrieval Quality Lab") ([GitHub](#))

Tech Stack: Python, ChromaDB (HNSW), Ollama, RAGAS, LangChain, BM25, sentence-transformers, Streamlit

- Built a **RAG** evaluation pipeline comparing BM25, dense, hybrid, and reranked retrieval methods
- Achieved **85.3% Recall@1** with hybrid retrieval plus cross-encoder reranking, up from 69.6% BM25
- Evaluated generation quality using EM, F1, a custom grounding score, and **RAGAS** relevancy
- Diagnosed generation, not retrieval, as the system bottleneck despite 95% gold-doc accuracy
- Built a 5-tab **Streamlit** dashboard visualizing retrieval ablations, generation quality, and latency
- Ran a local **Ollama** LLM for generation and LLM-as-judge evaluation with no paid APIs

Banking Liquidity Forecast Lab ([GitHub](#))

Tech Stack: Python, LightGBM, statsmodels (Holt-Winters), FastAPI, Docker Compose

- Built a retail-banking cash-outflow forecasting pipeline comparing seasonal naive, **Holt-Winters**, and **LightGBM**
- Achieved **9.48% WAPE** forecasting daily cash outflow under six-fold walk-forward validation
- Built a residual-based **anomaly detector** using robust z-scores, reaching 0.71 F1 at threshold
- Measured a marketing campaign's **causal lift** using interrupted time series and difference-in-differences
- Recovered a known **12% campaign lift** within 0.4 percentage points using two causal estimators
- Served the forecasting model through **FastAPI** endpoints packaged in reproducible Docker Compose containers

Conditional Generative Data Augmentation for Image Classification ([GitHub](#))

Tech Stack: Python, PyTorch, Conditional GAN (cGAN), Conditional VAE-GAN, CNN, MNIST, Fashion-MNIST

- Developed **CGAN** and Conditional **VAE-GAN** pipelines in PyTorch across MNIST and Fashion-MNIST
- Improved Fashion-MNIST accuracy from **63.6% to 70.8%** using only 100 real samples
- Evaluated seven classifier training strategies, finding weighted real-synthetic mixing performed best overall
- Ran architecture and training-strategy ablations, finding learning-rate ratio the strongest performance lever
- Found VAE-GAN augmentation underperformed real-only training, showing complexity alone doesn't improve results

TinyACE: Bounded Working-Memory Context Engineering for SLMs ([GitHub](#))

Tech Stack: Python, PyTorch, TinyLlama-1.1B, Phi-3 Mini (3.8B), Mistral-7B, sentence-transformers

- Developed **TINYACE**, a bounded working-memory framework capping context at 512 tokens for small models
- Improved **Phi-3 Mini** accuracy from 74% to 78% across a 50-question SciQ benchmark
- Ran a multi-scale evaluation across TinyLlama, Phi-3 Mini, and Mistral on one GPU
- Identified failure tracking as the most critical component through systematic ablation studies
- Found simple **FIFO** eviction matched or beat complex utility scoring for small models

Advanced Time Series Analysis & Predictive Dashboarding

Tech Stack: Python, Docker, Git, Statistical Trend Analysis

- Built a refreshable ingestion pipeline monitoring rolling 24-hour windows of operational metrics
- Applied moving averages and trend signals to detect anomalies in high-frequency data
- Dockerized the workflow and used Git version control for reproducible execution across environments

TECHNICAL SKILLS

- **Data & SQL** SQL, MySQL, ETL, Data Pipelines, Data Cleaning, Snowflake, DuckDB, PostgreSQL, Informatica
- **Data Engineering & Orchestration** dbt, Apache Kafka (Redpanda), Apache Spark (incl. Structured Streaming), Apache Airflow, Terraform, Databricks (Delta Lake, Unity Catalog)
- **Streaming Data Reliability & Cloud (AWS)** TypeScript, Node.js, Amazon Kinesis, Amazon S3, AWS CDK, AWS Lambda, Ajv/JSON Schema, Vitest
- **LLM & RAG Engineering** RAG Architecture, Hybrid Search (BM25 + Dense + RRF), Cross-Encoder Reranking, ChromaDB, Ollama, RAGAS, LangChain, sentence-transformers, Hugging Face Datasets
- **Statistics & Methods** Statistical Inference, Experimental Design, A/B Testing, Causal Inference (Interrupted Time Series, Difference-in-Differences), Time-Series Forecasting (Holt-Winters, Walk-Forward Validation), Model Evaluation
- **ML & AI** PyTorch, scikit-learn, LightGBM, Gradient Boosting, statsmodels, SHAP, GANs, VAE-GAN, CNNs, Deep Learning, LLM Context Engineering
- **Analytics & BI** Power BI, DAX, Tableau, Grafana, Plotly, Streamlit, Data Visualization
- **Programming & Tools** Python, pandas, NumPy, PyArrow, R, Git, Bitbucket, Docker, Docker Compose, FastAPI, Jupyter, CI/CD
- **AI-Assisted Development** GitHub Copilot, Claude AI, Cursor, AI Debugging
- **Salesforce Ecosystem** Salesforce CRM, Apex, AutoRABIT